Preface

Programming languages come and go constantly, and very few languages in use today have roots going back more than a decade or so. Some examples are COBOL, which is still used quite heavily in mainframe environments; Java, which was born in the mid-1990s and has become one of the most popular programming languages; and C, which is still quite popular for operating systems and server development and for embedded systems. In the database arena, we have SQL, whose roots go all the way back to the 1970s.

SQL was initially created to be the language for generating, manipulating, and retrieving data from relational databases, which have been around for more than 40 years. Over the past decade or so, however, other data platforms such as Hadoop, Spark, and NoSQL have gained a great deal of traction, eating away at the relational database market. As will be discussed in the last few chapters of this book, however, the SQL language has been evolving to facilitate the retrieval of data from various platforms, regardless of whether the data is stored in tables, documents, or flat files.

Why Learn SQL?

Whether you will be using a relational database or not, if you are working in data science, business intelligence, or some other facet of data analysis, you will likely need to know SQL, along with other languages/platforms such as Python and R. Data is everywhere, in huge quantities, and arriving at a rapid pace, and people who can extract meaningful information from all this data are in big demand.

Why Use This Book to Do It?

There are plenty of books out there that treat you like a dummy, idiot, or some other flavor of simpleton, but these books tend to just skim the surface. At the other end of the spectrum are reference guides that detail every permutation of every statement in a language, which can be useful if you already have a good idea of what you want to do but just need the syntax. This book strives to find the middle ground, starting with some background of the SQL language, moving through the basics, and then progressing into

some of the more advanced features that will allow you to really shine. Additionally, this book ends with a chapter showing how to query data in nonrelational databases, which is a topic rarely covered in introductory books.

Structure of This Book

This book is divided into 18 chapters and 2 appendixes:

Chapter 1, "A Little Background"

Explores the history of computerized databases, including the rise of the relational model and the SQL language.

Chapter 2, "Creating and Populating a Database"

Demonstrates how to create a MySQL database, create the tables used for the examples in this book, and populate the tables with data.

Chapter 3, "Query Primer"

Introduces the `select` statement and further demonstrates the most common clauses (`select`, `from`, `where`).

Chapter 4, "Filtering"

Demonstrates the different types of conditions that can be used in the `where` clause of a `select`, `update`, or `delete` statement.

Chapter 5, "Querying Multiple Tables"

Shows how queries can utilize multiple tables via table joins.

Chapter 6, "Working with Sets"

This chapter is all about data sets and how they can interact within queries.

Chapter 7, "Data Generation, Manipulation, and Conversion"

Demonstrates several built-in functions used for manipulating or converting data.

Chapter 8, "Grouping and Aggregates"

Shows how data can be aggregated.

Chapter 9, "Subqueries"

Introduces subqueries (a personal favorite) and shows how and where they can be utilized.

Chapter 10, "Joins Revisited"

Further explores the various types of table joins.

Chapter 11, "Conditional Logic"

Explores how conditional logic (i.e., if-then-else) can be utilized in `select`, `insert`, `update`, and `delete` statements.

Chapter 12, "Transactions"

Introduces transactions and shows how to use them.

Chapter 13, "Indexes and Constraints"

Explores indexes and constraints.

Chapter 14, "Views"

Shows how to build an interface to shield users from data complexities.

Chapter 15, "Metadata"

Demonstrates the utility of the data dictionary.

Chapter 16, "Analytic Functions"

Covers functionality used to generate rankings, subtotals, and other values used heavily in reporting and analysis.

Chapter 17, "Working with Large Databases"

Demonstrates techniques for making very large databases easier to manage and traverse.

Chapter 18, "SQL and Big Data"

Explores the transformation of the SQL language to allow retrieval of data from nonrelational data platforms.

Appendix A, "ER Diagram for Example Database"

Shows the database schema used for all examples in the book.

Appendix B, "Solutions to Exercises"

Shows solutions to the chapter exercises.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

`Constant width`

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

`Constant width italic`

Shows text that should be replaced with user-supplied values or by values determined by context.

Constant width bold

Shows commands or other text that should be typed literally by the user.

NOTE

Indicates a tip, suggestion, or general note. For example, I use notes to point you to useful new features in Oracle9*i*.

WARNING

Indicates a warning or caution. For example, I'll tell you if a certain SQL clause might have unintended consequences if not used carefully.

Using the Examples in This Book

To experiment with the data used for the examples in this book, you have two options:

- Download and install the MySQL server version 8.0 (or later) and load the Sakila example database from *https://dev.mysql.com/doc/index-other.html*.
- Go to *https://learning.oreilly.com/scenarios/mysql-sandbox/9781492079705* to access the MySQL Sandbox, which has the Sakila Sample database loaded in a MySQL instance. If you don't have one already, you'll have to set up an O'Reilly account. Then, click on the Start button and begin.

If you choose the second option, once you start the scenario, a MySQL server is installed and started, and then the Sakila schema and data are loaded. When it's ready, a standard `mysql>` prompt appears, and you can then start querying the sample database. This is certainly the easiest option, and I anticipate that most readers will choose this option; if this sounds good to you, feel free to skip ahead to the next section.

If you prefer to have your own copy of the data and want any changes you have made to be permanent, or if you are just interested in installing the MySQL server on your own machine, you may prefer the first option. You may also opt to use a MySQL server hosted in an environment such as Amazon Web Services or Google Cloud. In either case, you will need to perform the installation/configuration yourself, as it is beyond the scope of this

book. Once your database is available, you will need to follow a few steps to load the Sakila sample database.

First, you will need to launch the `mysql` command-line client and provide a password, and then perform the following steps:

1. Go to *https://dev.mysql.com/doc/index-other.html* and download the files for "sakila database" under the Example Databases section.
2. Put the files in a local directory such as *C:\temp\sakila-db* (used for the next two steps, but overwrite with your directory path).
3. Type `source c:\temp\sakila-db\sakila-schema.sql;` and press Enter.
4. Type `source c:\temp\sakila-db\sakila-data.sql;` and press Enter.

You should now have a working database populated with all the data needed for the examples in this book.

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NOTE

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1. A Little Background ›